

Aliah University

End-Semester (autumn) Examination - 2021

(BTech, CSE 5th Semester)

Paper Name: Formal Languages and Automata Theory

Full Marks: 80

Paper Code: CSEUGPC13/CSE307 (Regular/Supplementary)

Time: 3 Hrs

Group – A

1. Choose correct alternative –

(10X2=20)

I. Which of the following is a regular language?

(A) $L = a^{2^n} \mid n \geq 1$

(B) $L = a^{n!} \mid 1 \leq n \leq 1000$

(C) $L = a^p \mid p \text{ is prime}$

(D) $L = xx \mid x \in a, b^*$

II. Which of the following defines string “w” of language that can be generated by the following CFG?

$S \rightarrow XY$

$X \rightarrow aX \mid bX \mid a$

$Y \rightarrow Ya \mid Yb \mid a$

(A) w has atleast one ‘b’

(B) w should end with ‘a’

(C) w has no consecutive ‘a’s or ‘b’s

(D) w has atleast two consecutive ‘a’s

III. Consider the pair of regular expressions given below:

(I) $0^*(10^*)^* \& (1^*0)^*1^*$

(II) $(r^*+s^*) \& (r+s)^*$

(III) $(a^*+b)^* \& (a+b^*)^*$

(IV) $(PQ)^*P \& P(QP)^*$

Which of the above pairs represent equivalent regular expressions?

(A) (II) and (I) only

(B) (III) & (IV) only

(C) (I), (III), (IV) only

(D) All of these

IV. The number of states in minimal DFA which accepts all the strings that contain at least two 0’s and at most one 1 over the alphabet = {0, 1} is –

(A) 4

(B) 5

(C) 6

(D) 7

V. Which of the following is context free language?

$L_1 = \text{complement of } \{a^i b^j c^k \mid i \neq j \text{ and } j \neq k\}$

$L_2 = \{ww \mid w \in (0,1)^*\}$

$L_3 = \{1^{n!} \mid n \geq 0\}$

$L_4 = \{0^n 1^n 0^n 1^n \mid n \geq 0\}$

(A) L_1 only

(B) L_1 and L_2 only

(C) L_1 and L_4 only

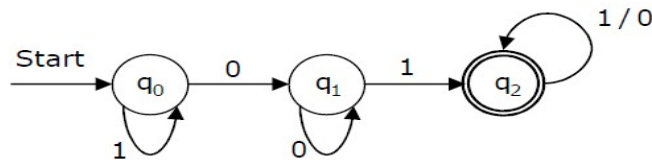
(D) none of the given languages

- VI. What will be the number of states in the minimal DFA which accepts the language represented by regular expression $(a + aaa)^*$ over the alphabet $\{a\}$?
 (A) 1 (B) 2 (C) 3 (D) 4

- VII. Match the correct automaton given in Y for the given transition function in X.

X	Y
I. $Q \times \Sigma \rightarrow Q$	A. NFA
II. $Q \times \Sigma \rightarrow 2^Q$	B. DFA
III. $Q \times \Sigma \rightarrow Q \times \{L, R\}$	C. NFA with null moves
IV. $Q \times \{\Sigma \cup \epsilon\} \rightarrow 2^Q$	D. Turning machine
	E. 2 way DFA
	F. PDA
(A) I-B, II-A, III-D, IV-F	(B) I-B, II-A, III-E, IV-C
(C) I-A, II-B, III-E, IV-F	(D) I-B, II-A, III-F, IV-D

- VIII. Consider the following DFA.



What is the language accepted by above DFA?

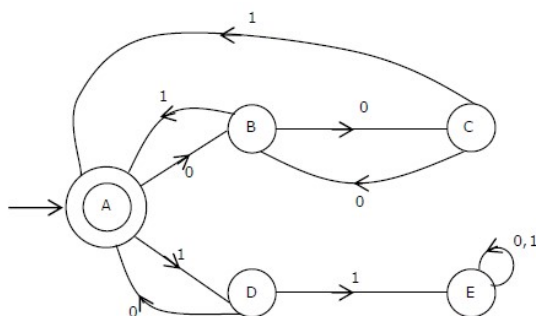
- (A) Strings over $\Sigma = \{0, 1\}$ contains '01' as a substring.
 (B) Strings over $\Sigma = \{0, 1\}$ contains '01' as ending string.
 (C) Strings over $\Sigma = \{0, 1\}$ contains '01' as starting string.
 (D) None of these
- IX. Which of the following language is regular?
 (A) $\{a^{2n}b^m c^n \mid n, m \geq 0\}$ (B) $\{a^n b^{2m} c^k \mid n, m, k \geq 0\}$
 (C) $\{b^m c^{2k} a^n \mid n = m, k \geq 0\}$ (D) None of these
- X. Which of the following languages are DCFL?
 $L_1 = \{WW^R \mid W \in \{a, b\}^* \text{ and } W^R \text{ is reverse of } W\}$
 $L_2 = \{WW^R x \mid W, x \in \{0, 1\}^*\}$
 (A) L_1 only (B) L_2 only
 (C) Both L_1 and L_2 (D) Neither of L_1 and L_2

Group - B
(Answer any 4 questions)

(4X5=20)

- Construct a Deterministic Finite Automata for the language given below –
***L*: All strings in which last two symbols are same, over alphabet $\Sigma = \{0, 1\}$**
- Using Pumping Lemma prove that –
***L*: $\{ww \mid w \text{ means strings formed over } \Sigma = \{a, b\}\}$ is not a Context Free Language.**

4. Using Arden's theorem find Regular Expression for the Finite Automata given below-



5. Consider **regular expression of a Language L: $(a + b)^* aa (a + b)^+$**
 a) Right down Left-Linear Regular Grammar.
 b) What will be number of states in the minimized DFA which accepts L?
6. Discuss closure properties of Regular Language and Deterministic Context free Language under operations – Union, Intersection, Compliment, Concatenation and Reverse.

Group – C
(Answer any 4 questions)

(10X4=40)

7. Consider the language below to answer the following questions –
L: All strings, either starts with '00' or ends with '11', over alphabet $\Sigma = \{0, 1\}$
 a) Construct ϵ -NFA which accepts L.
 b) Convert the ϵ -NFA into DFA. Perform minimization if required.
 c) Write down Regular grammar which generates L. 3+3+1+3=10
8. A Context free language is given below -
L: $\{a^n b^m \mid n \neq m, \text{ where } \Sigma = \{a, b\}\}$
 a) Design a PDA to accept all valid strings of L.
 b) Write down the Grammar which generates L. (Mention all 4tuples) 5X2=10
9. a) Construct a Turing Machine to accept the language given below -
L: $\{wcw \mid \text{where 'w' means strings formed over inputs } \{a, b\} \text{ and } c \text{ is a terminal}\}$
 b) Explain Halting Problem with an example. 8+2=10
10. a) Write down differences between DFA and NFA. 3+3+4=10
 b) Compare Recursive and Recursive enumerable languages.
 c) Design a Moore machine to determine the residue of mod 2 of the input treated as a binary string.
11. a) Consider the grammar below and answer. 2+4+4=10
P: $E \rightarrow E + E \mid E * E \mid a \mid b \mid c$ where $N = \{E\}$, $T = \{a, b, c, +, *\}$ and E is the starting symbol.
 Perform LMD to generate a string " $a*b+c$ ". Is it an ambiguous grammar? Justify your answer.
 b) Write the Grammar for the language below –
L = $\{a^n b^m c^{n+m} \mid n, m \geq 1 \text{ and } \Sigma = \{a, b, c\}\}$

END